

Ali Momennasab

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Education

University of California, San Diego

8/2026 - 5/2028 (expected)

Master of Science, Computer Science | Deep Learning and Computer Vision Focus

California State Polytechnic University, Pomona

8/2022 - 1/2026

Bachelor of Science, Computer Science, Minor in Data Science | **3.93/4.0 GPA**

Pomona, CA

- Relevant Coursework: Computer Vision, Cloud Computing, GPU Computing, Generative AI

Experience

Research Assistant | University of California, San Francisco

6/2023 - Present

Researching deep learning segmentation of cardiac ultrasounds under Professor Jan Christoph.

- Developed a PyTorch temporal U-Net for cardiac ultrasound segmentation with residual and dense block variants and on-the-fly 3D augmentations, improving F1 from 0.79 (previous best model) to 0.92.
- Trained conditional GANs and diffusion models to generate synthetic ultrasound volumes from segmentation masks, achieving average Dice scores of 0.88 between segmented synthetic volumes and masks.

Research Assistant | California State Polytechnic University, Pomona

1/2025 - Present

Researching deep learning survival analysis of whole-slide cancer images under Professor Sai Kosaraju.

- Built a multimodal data pipeline integrating pathology images and genomic features for a CNN survival model, achieving C-index of 0.95 (brain) and 0.73 (lung). Presented at Cal Poly Pomona research symposium.
- Developed a hybrid Vision Transformer and GNN framework for whole-slide image embedding and tumor subtype clustering, achieving F1 score of 0.82.

Instructor | Code Ninjas Hacienda Heights

6/2023 - 8/2024

- Delivered 1-on-1 coding lessons to K-12 students in Unity, Roblox Studio, and Microsoft MakeCode.
- Led weekly web development summer camps, co-developing an HTML, CSS, and JavaScript curriculum.

Projects

ASL Word & Letter Translator | github.com/alimomennasab/ASL-Translator

- Developed and trained 2D CNNs for static ASL letter classification (95% accuracy), and 3D CNNs for video-based word translation (40% accuracy vs. 65% SOTA on benchmark dataset).
- Built a MediaPipe hand detection pipeline to crop and augment (flips, rotations, grayscale, speed adjustments) signing regions, expanding dataset from 500 to 5,000 samples and improving accuracy from 20% to 40%.

Generative Music Genre Transfer | github.com/alimomennasab/CS4990-Generative-AI

- Implemented VAE, GAN, and WGAN models in Pytorch for symbolic music genre transfer, observed greater training stability and reconstruction consistency with VAEs.
- Designed a MIDI file tokenization pipeline, converting 2,220 jazz, pop, and classical songs into 13 million tokens for model training.

Content Moderator Agent | github.com/alimomennasab/content-moderation-agent

- Built a content moderation agent using GPT-4o-mini to route text inputs to hate speech (RoBERTa) or spam (BERT) classifiers, reducing API calls by pre-filtering content.
- Developed a Netlify-deployed full-stack app with a Next.js frontend, FastAPI backend, and PostgreSQL database, integrating OpenAI and HuggingFace APIs.

Skills

Languages: Java, C/C++/CUDA C, Python, TypeScript/JavaScript, MATLAB

Frameworks & Tools: PyTorch, TensorFlow, NumPy, Scikit-learn, React, Next.js, Node.js, Express, Git, Figma

Databases & Cloud: MongoDB, PostgreSQL, Prisma, AWS, Pinecone